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Battery module and battery pack including the same

Abstract

A battery module includes a battery cell stack in which a plurality of battery cells are stacked, a bus bar frame coupled to each of the front and rear ends of the battery cell stack, a frame member that houses a cell block including the battery cell stack and the bus bar frame, a pad part located at one end of the bottom part of the frame member, and a film part connected to the pad part and protruding toward the bus bar frame.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0190264	12/2016	Kim et al.	N/A	N/A
2018/0138565	12/2017	Lee et al.	N/A	N/A
2019/0074557	12/2018	Shin et al.	N/A	N/A
2019/0280285	12/2018	Jansen et al.	N/A	N/A
2020/0194851	12/2019	Sec et al.	N/A	N/A
2020/0395643	12/2019	Seo et al.	N/A	N/A
2021/0228291	12/2020	Kim et al.	N/A	N/A
2022/0052391	12/2021	Kim et al.	N/A	N/A
2022/0102813	12/2021	Cho et al.	N/A	N/A
2022/0384875	12/2021	Park	N/A	H01M 10/647
2022/0416327	12/2021	Jang	N/A	H01M 50/507

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108140793	12/2017	CN	N/A
109216610	12/2018	CN	N/A
209312824	12/2018	CN	N/A
210349920	12/2019	CN	N/A
3 113 278	12/2017	EP	N/A
3 913 702	12/2020	EP	N/A
10-2019-0139620	12/2012	KR	N/A
10-1916720	12/2017	KR	N/A
10-2019-0110782	12/2018	KR	N/A
10-2050530	12/2018	KR	N/A

10-2020-0030967	12/2019	KR	N/A
10-2020-0068478	12/2019	KR	N/A
10-2020-0109127	12/2019	KR	N/A
10-2020-0142240	12/2019	KR	N/A
10-2020-0142242	12/2019	KR	N/A
2018/022907	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report (PCT/ISA/210) issued in PCT/KR2021/013753 mailed on Feb. 8, 2022. cited by applicant

Extended European Search Report for European Application No. 218833069.3, dated Jul. 31, 2023. cited by applicant

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Background/Summary

TECHNICAL FIELD

Cross Citation with Related Application(s)

(1) This application claims the benefit of Korean Patent Application No. 10-2020-0135020 filed on Oct. 19, 2020 and Korean Patent Application No. 10-2020-0183858 filed on Dec. 24, 2020 with the Korean Intellectual Property Office, the disclosures of which are incorporated herein by reference in their entirety.

(2) The present disclosure relates to a battery module and a battery pack including the same, and more particularly to a battery module which improves the assembly property and insulation property, and a battery pack including the same.

BACKGROUND

(3) Secondary batteries, which are easily applicable to various product groups and has electrical characteristics such as high energy density, are universally applied not only for a portable device but also for an electric vehicle or a hybrid electric vehicle, an energy storage system or the like, which is driven by an electric driving source. Such secondary battery is attracting attention as a new environment-friendly energy source for improving energy efficiency since it gives a primary advantage of remarkably reducing the use of fossil fuels and also does not generate by-products from the use of energy at all.

(4) Small-sized mobile devices use one or several battery cells for each device, whereas middle- or large-sized devices such as vehicles require high power and large capacity. Therefore, a middle or large-sized battery module having a plurality of battery cells electrically connected to one another is used.

(5) The middle- or large-sized battery module is preferably manufactured so as to have as small a size and weight as possible. Therefore, a prismatic battery, a pouch type battery or the like, which can be stacked with high integration and has a small weight relative to capacity, is usually used as a battery cell of the middle- or large-sized battery module. Meanwhile, in order to protect the battery cell stack from external impact, heat or vibration, the battery module may include a module frame that is opened in its front and rear surfaces and houses the battery cell stack in an internal space.

(6) FIG. 1 is an exploded perspective view illustrating a battery module having a mono frame according to the related art.

(7) Referring to FIG. 1, a battery module may include a battery cell stack **12** formed by stacking a plurality of battery cells **11**, a mono frame **20** of which a front surface and a rear surface are opened so as to cover the battery cell stack **12**, and end plates **60** that cover the front and rear surfaces of the mono frame **20**. In order to form such a battery module, it is necessary to horizontally assemble the battery module such that the battery cell stack **12** is inserted into the opened front or rear surface of the mono frame **20** along the X-axis direction as shown by the arrow in FIG. 1.

However, in order to stably perform such a horizontal assembly, a sufficient clearance has to be secured between the battery cell stack **12** and the mono frame **20**. Here, the clearance refers to a gap generated by press-fitting and the like.

(8) A thermal conductive resin layer (not shown) may be formed between the battery cell stack **12** and the mono frame **20**. The thermal conductive resin layer can play a role of transferring the heat generated from the battery cell stack to the outside of the battery module, and fixing the battery cell stack inside the battery module. When the clearance becomes larger, the use amount of the thermal conductive resin layer may become larger than necessary.

(9) In addition, the height of the mono frame **20** should be designed large in consideration of the maximum height of the battery cell stack **12** and an assembly tolerance during the insertion process, and the like, which may lead to generation of unnecessary wasted space.

(10) Further, there is a need to reinforce an insulation problem that may occur between the battery cell stack **12** and a frame member for housing the battery cell stack **12** as in the mono frame **20**.

DETAILED DESCRIPTION OF THE INVENTION

Technical Problem

(11) It is an object of the present disclosure to provide a battery module which improves the assembly property and insulation property, and a battery pack including the same.

(12) However, the technical problem to be solved by embodiments of the present disclosure is not limited to the above-described problems, and can be variously expanded within the scope of the technical idea included in the present disclosure.

Technical Solution

(13) According to one embodiment of the present disclosure, there is provided a battery module comprising: a battery cell stack in which a plurality of battery cells are stacked, a bus bar frame coupled to at least one of the front and rear ends of the battery cell stack, a frame member that houses a cell block, the cell block including the battery cell stack and the bus bar frame, a pad part located at one end of the bottom part of the frame member, and a film part connected to the pad part and protruding toward the bus bar frame.

(14) A bottom part of the frame member comprises a first portion and a second portion, the first portion located at the edge with respect to a longitudinal direction of the battery cell, the second portion located inside the first portion, and a thickness of the first portion may be less than a thickness of the second portion.

(15) The battery further comprises a thermal conductive resin layer located between the second portion and the battery cell stack, wherein the pad part may be located between the thermal conductive resin layer and the first portion.

(16) The pad part may be located on the second portion, and the film part is located on the first portion.

(17) The film part may be formed so as to be in contact with a step portion connecting the first portion and the second portion of the bottom part of the frame member.

(18) The frame member is opened on both sides facing each other with respect to the longitudinal direction of the battery cell, the bus bar frame is connected to the battery cell stack on the opened both sides of the frame member, the bus bar frame comprises a main frame arranged perpendicular to the longitudinal direction of the battery cell and a bending part extending from a lower part of the main frame, and the bending part may be located on the first portion of the bottom part of the frame member.

- (19) The film part may extend between the bending part of the bus bar frame and the second portion of the bottom part of the frame member.
- (20) The film part may further comprise a protrusion film part located between the bending part of the bus bar frame and the first portion of the bottom part of the frame member.
- (21) The sum of a thickness of the bending part and a thickness of the first portion may be less than a thickness of the second portion.
- (22) The battery cell comprises a protrusion part formed in a width direction, and the protrusion part may be located on the bending part.
- (23) The pad part and the film part may be integrally formed.
- (24) The film part may be attached to a lower end of the pad part.
- (25) The battery module further comprises end plates each coupled to opened sides of the frame member, wherein the opened sides of the frame member may face each other with respect to a longitudinal direction of the battery cell.
- (26) The frame member may comprise a module frame that houses the cell block and has an opened upper part, and an upper plate that covers the cell block on the module frame, and the module frame may comprise a bottom part of the frame member and two side surface parts facing each other.
- (27) According to another embodiment of the present disclosure, there is provided a battery pack comprising the above-mentioned battery module.

Advantageous Effects

- (28) According to embodiments of the present disclosure, by modifying the structure of the existing mono frame, it is possible to reduce the tolerance between the battery cell stack and the frame member and improve the space utilization rate, as compared with a conventional technique.
- (29) Further, by utilizing the pad, which is an overflow prevention structure, it is possible to prevent the thermal conductive resin from flowing into an unintended space when the cell block is inserted.
- (30) Additionally, by integrally forming the pad and the insulating film, which are the overflow prevention structures, it is possible to reinforce the insulating performance between the battery cell and the module frame while simplifying the assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an exploded perspective view illustrating a battery module having a mono frame according to the related art;
- (2) FIG. 2 is an exploded perspective view illustrating a battery module according to an embodiment of the present disclosure;
- (3) FIG. 3 is a perspective view illustrating a state in which components constituting the battery module of FIG. 2 are combined;
- (4) FIG. 4 is a perspective view illustrating one battery cell included in the battery cell stack of FIG. 2;
- (5) FIG. 5 is a perspective view illustrating a bus bar frame in the battery module of FIG. 2;
- (6) FIG. 6 is a cross-sectional view taken along the xz plane which is the longitudinal direction of the battery cell stack in FIG. 3;
- (7) FIG. 7 is a perspective view illustrating a module frame in the battery module of FIG. 2;
- (8) FIG. 8 is a perspective view illustrating a state in which a cell block is inserted into the module frame of FIG. 7;
- (9) FIG. 9 is a cross-sectional view of a battery module corresponding to the comparative example of FIG. 6;

(10) FIG. 10 is a modified embodiment of FIG. 6, which is a cross-sectional view taken along the xz plane which is the longitudinal direction of the battery cell stack in FIG. 3;
(11) FIG. 11 is a perspective view of a battery module including the film part of FIG. 10; and
(12) FIG. 12 is a modified embodiment of FIG. 6, which is a cross-sectional view taken along the xz plane which is the longitudinal direction of the battery cell stack in FIG. 3.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(13) Hereinafter, various embodiments of the present disclosure will be described in detail with reference to the accompanying drawings so that those skilled in the art can easily carry out them. The present disclosure may be modified in various different ways, and is not limited to the embodiments set forth herein.

(14) Portions that are irrelevant to the description will be omitted to clearly describe the present disclosure, and like reference numerals designate like elements throughout the specification.

(15) Further, in the drawings, the size and thickness of each element are arbitrarily illustrated for convenience of the description, and the present disclosure is not necessarily limited to those illustrated in the drawings. In the drawings, the thickness of layers, regions, etc. are exaggerated for clarity. In the drawings, for convenience of the description, the thicknesses of some layers and regions are shown to be exaggerated.

(16) In addition, it will be understood that when an element such as a layer, film, region, or plate is referred to as being “on” or “above” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, it means that other intervening elements are not present. Further, the word “on” or “above” means disposed on or below a reference portion, and does not necessarily mean being disposed “on” or “above” the reference portion toward the opposite direction of gravity.

(17) Further, throughout the specification, when a portion is referred to as “including” a certain component, it means that the portion can further include other components, without excluding the other components, unless otherwise stated.

(18) Further, throughout the specification, when referred to as “planar”, it means when a target portion is viewed from the upper side, and when referred to as “cross-sectional”, it means when a target portion is viewed from the side of a cross section cut vertically.

(19) FIG. 2 is an exploded perspective view illustrating a battery module according to an embodiment of the present disclosure. FIG. 3 is a perspective view illustrating a state in which components constituting the battery module of FIG. 2 are combined. FIG. 4 is a perspective view illustrating one battery cell included in the battery cell stack of FIG. 2.

(20) Referring to FIGS. 2 and 3, a battery module **100** according to the present embodiment includes a battery cell stack **120** containing a plurality of battery cells **110**, a bus bar frame **130** coupled to each of the front and rear ends of the battery cell stack **120**, and a frame member **500** that houses a cell block **140** including the battery cell stack **120** and the bus bar frame **130**. The frame member **500** may include a module frame **300** in which the cell block **140** is mounted and the upper part is opened, an upper plate **400** that covers the cell block **140** at the upper part of the module frame **300**. The module frame **300** includes two side surface parts facing each other and a bottom part connecting the two side surface parts. The two side surface parts and the bottom part may be integrally formed. The module frame **300** may be U-shaped.

(21) The frame member **500** may be opened on both sides facing each other with respect to the direction in which the electrode leads of the battery cell stack **120** protrude. The battery module **100** according to the present embodiment may further include an end plate **150** coupled to the cell block on opened both sides of the frame member **500**.

(22) The battery module **100** according to the present embodiment includes a thermal conductive resin layer **310** located between the module frame **300** and the battery cell stack **120**. The thermal conductive resin layer **310** is a kind of heat dissipation layer, and may be formed by applying a

material having a heat dissipation function. The end plate **150** may be formed of a metal material. (23) When opened both sides of the module frame **300** are referred to as a first side and a second side, respectively, the module frame **300** has a plate-shaped structure that is bent so as to continuously warp the left side, lower and right side surfaces adjacent to each other among the remaining outer surfaces excluding surfaces of the cell block **140** corresponding to the first side and the second side. The upper part corresponding to the bottom part of the module frame **300** is opened.

(24) The upper plate **400** has a single plate-shaped structure that covers the remaining upper part excluding the left side, lower and right side surfaces which are wrapped by the module frame **300**. The module frame **300** and the upper plate **400** can be coupled by welding or the like in a state in which the corresponding edge areas are in contact with each other, thereby forming a structure wrapping the cell block **140**. That is, the module frame **300** and the upper plate **400** can have a coupling part CP formed at an edge area corresponding to each other by a coupling method such as welding.

(25) The battery cell stack **120** includes a plurality of battery cells **110** stacked in one direction, and the plurality of battery cells **110** may be stacked in the y-axis direction as shown in FIG. 2. In other words, a direction in which the plurality of battery cells **110** are stacked may be the same as a direction in which two side surface parts of the module frame **300** face each other.

(26) The battery cell **110** is preferably a pouch type battery cell. For example, referring to FIG. 4, the battery cell **110** according to the present embodiment may have a structure in which the two electrode leads **111** and **112** protrude from one end part **114a** and the other end part **114b** of the battery body **113** toward mutually opposite directions, respectively. The battery cell **110** can be manufactured by joining both end parts **114a** and **114b** of the cell case **114** and both side surfaces **114c** connecting them in a state in which an electrode assembly (not shown) is housed in the cell case **114**. In other words, the battery cell **110** according to the present embodiment has a total of three sealing parts **114sa**, **114sb** and **114sc**, wherein the sealing parts **114sa**, **114sb** and **114sc** have a structure that is sealed by a method such as heat fusion, and the remaining other side part may be formed of a connection part **115**. A space between both end parts **114a** and **114b** of the battery case **114** is defined as a longitudinal direction of the battery cell **110**, and a space between the one side surface **114c** and the connection part **115** that connect both end parts **114a** and **114b** of the battery case **114** is defined as a width direction of the battery cell **110**.

(27) The connection part **115** is a region extending long along one edge of the battery cell **110**, and a protrusion part **110p** of the battery cell **110** may be formed at an end part of the connection part **115**. The protrusion part **110p** may be formed on at least one of both end parts of the connection part **115** and may protrude in a direction perpendicular to the direction in which the connection part **115** extends. The protrusion part **110p** may be located between one of the sealing parts **114sa** and **114sb** of both end parts **114a** and **114b** of the battery case **114**, and the connection part **115**.

(28) The cell case **114** is generally formed of a laminate structure of a resin layer/metallic thin film layer/resin layer. For example, a surface of the battery case formed of an O(oriented)-nylon layer tends to slide easily by an external impact when a plurality of battery cells are stacked in order to form a medium- or large-sized battery module. Therefore, in order to prevent this sliding and maintain a stable stacked structure of the battery cells, an adhesive member, for example, a sticky adhesive such as a double-sided tape or a chemical adhesive coupled by a chemical reaction upon adhesion, can be attached to the surface of the battery case to form the battery cell stack **120**. In the present embodiment, the battery cell stack **120** is stacked in a Y-axis direction and housed into the module frame **300** in a Z-axis direction, and then can be cooled by a thermal conductive resin layer described later. As a comparative example thereto, there is a case in which the battery cells are formed as cartridge-shaped components so that fixing between the battery cells is made by assembling the battery module frame. In this comparative example, due to the presence of the cartridge-shaped components, there is almost no cooling action or the cooling may be proceeded in

a surface direction of the battery cells, whereby the cooling does not well perform toward a height of the battery module.

(29) Referring to FIGS. 2 and 7, the module frame **300** according to the present embodiment includes a bottom part **300a** and two side surface parts **300b** facing each other connected by the bottom part **300a**. Before the battery cell stack **120** is mounted on the bottom part **300a** of the module frame **300**, a thermal conductive resin is applied to the bottom part of the module frame **300**, and the thermal conductive resin can be cured to form a thermal conductive resin layer **310**. The thermal conductive resin layer **310** is located between the bottom part **300a** of the module frame **300** and the battery cell stack **120**, and can serve to transfer heat generated in the battery cell **110** to the bottom of the battery module **100** and fix the battery cell stack **120**.

(30) According to the present embodiment, the thermal conductive resin layer **310** includes a plurality of application lines that extend long in a direction perpendicular to the direction in which the plurality of battery cells **110** are stacked. The plurality of application lines may form two groups, and an insulating film **330** may be formed between the two groups. The insulating film **330** can function to maintain insulating performance between the battery cell **110** and the module frame **300**, and at least a part of a thermal conductive resin may be applied onto the insulating film **330**.

(31) According to the present embodiment, as shown in FIG. 2, an insulation reinforcing member **345** for reinforcing insulation performance between the battery cell stack **120** and the frame member **500** is formed on the bottom part of the module frame **300**. The insulation reinforcing member **345** will be described in detail below.

(32) FIG. 5 is a perspective view illustrating a bus bar frame in the battery module of FIG. 2.

(33) Referring to FIG. 5, the bus bar frame according to the present embodiment comprises a main frame **130a** arranged perpendicular to a direction in which the electrode leads **111** and **112** protrude and a bending part **130b** extending from a lower part of the main frame **130a**. The bus bar frame **130** is connected to the battery cell stack **120** as described with reference to FIGS. 2 and 3. A structure in which the electrode leads pass through slits and are coupled to bus bars may be formed in the main frame **130a**. The bending part **130b** may be bent by about 90 degrees with respect to the main frame **130a** and may be located on the bottom part **300a** of FIG. 2. The bending part **130b** and peripheral configurations will be additionally described with reference to FIG. 6.

(34) FIG. 6 is a cross-sectional view taken along the xz plane which is the longitudinal direction of the battery cell stack in FIG. 3. FIG. 7 is a perspective view illustrating a module frame in the battery module of FIG. 2. FIG. 8 is a perspective view illustrating a state in which a cell block is inserted into the module frame of FIG. 7.

(35) Referring to FIG. 6, the battery cell **110** according to the present embodiment includes a protrusion part **110p** formed in a width direction, and the protrusion part **110p** is located on the bending part **130b**. Here, the width direction of the battery cell **110** may be the z-axis direction of FIG. 6. The bottom part **300a** of the module frame **300** according to the present embodiment includes a first portion **300a1** and a second portion **300a2**, wherein the first portion **300a1** is located at an edge based on the longitudinal direction of the battery cells **110**, and the second portion **300a2** is located inside the first portion **300a1**. At this time, it is preferable that the thickness of the first portion **300a1** is thinner than the thickness of the second portion **300a2**. Here, the longitudinal direction of the battery cell **110** may be the x-axis direction of FIG. 6.

(36) Referring to FIGS. 5 and 6, the bending part **130b** of the bus bar frame **130** according to the present embodiment is located on the first portion **300a1** of the bottom part **300a** of the module frame. At this time, it is preferable that the sum of the thickness of the bending part **130b** and the thickness of the first portion **300a1** is thinner than the thickness of the second portion **300a2**. This is because the protrusion part **110p** of the battery cell **110** is caught by a step between the first portion **300a1** and the second portion **300a2** and thus prevents the battery cell from flowing due to an external impact. In addition, a gap between the battery cell **110** and the frame member may be reduced through the processing of the bottom part **300a** of the module frame, and this gap

reduction effect may cause synergy with a gap reduction effect that may be obtained through height-direction assembling, thereby maximizing the overall space efficiency. The processing of the bottom part **300a** of the module frame may be performed simultaneously while forming the module frame structure. Press forming, NC (numerical control work) processing, or the like may be used in order to form such a step.

(37) The pad part **320** is located between the second portion **300a2** of the bottom part **300a** and the battery cell **110**, and the thermal conductive resin layer **310** is located inside the pad part **320**. That is, the pad part **320** may be located between the thermal conductive resin layer **310** and the first portion **300a1** of the bottom part **300a** to define a position where the thermal conductive resin layer **310** is formed. The battery module according to the present embodiment includes a film part **340** connected to the pad part **320** and protruding toward the bus bar frame **130**. The pad part **320** may be located on the second portion **300a2**, and the film part **340** may be located on the first portion **300a1**. At this time, the pad part **320** and the film part **340** may be integrally formed so as to constitute an insulation reinforcing member **345**. The film part **340** not only reinforces the insulation between the battery cell **110** or the protrusion part **110p** of the battery cell **110** and the module frame **300**, but is also integrally formed with the pad part **320**, thereby simplifying the assembly with the module frame. In a modified embodiment, the film part **340** may be attached to the lower end of the pad part **320**. Both the pad part **320** and the film part **340** are insulating parts, and may be flexible and compressively deformable members. In one example, the pad part **320** may be formed of polyurethane, and the film part **340** may be formed of polyethylene terephthalate (PET). Alternatively, both the pad part **320** and the film part **340** may be formed of polyurethane or polyethylene terephthalate.

(38) Referring to FIGS. 7 and 8, before the cell block **140** is mounted on the module frame **300**, an insulation reinforcing member **345** may be formed on the bottom part **300a** of the module frame **300**.

(39) FIG. 9 is a cross-sectional view of a battery module corresponding to the comparative example of FIG. 6.

(40) Referring to FIG. 9, the insulation reinforcing member according to the comparative example can form the insulating film part **340'** in a form separated from the pad part **320**. The insulating film part **340'** may be formed between the second portion **300a2** of the bottom part **300a** of the module frame and the bending part **130b** of the bus bar frame. According to the comparative example, alignment is required through an attachment process to form the insulating film part **340'**, and it is difficult to guarantee the insulation of the exposed portion between the insulating film part **340'** and the pad part **320** and/or between the insulating film part **340'** and the bending part **130b**.

(41) FIG. 10 is a modified embodiment of FIG. 6, which is a cross-sectional view taken along the xz plane which is the longitudinal direction of the battery cell stack in FIG. 3. FIG. 11 is a perspective view of a battery module including the film part of FIG. 10.

(42) Referring to FIGS. 10 and 11, the film part **340** may further include a protruding film part **340p** located between the bending part **130b** of the bus bar frame and the first portion **300a1** of the bottom part **300a** of the frame member. The protruding film part **340p** may vertically overlap the bending part **130b** of the bus bar frame. Without such an overlapping structure, a gap is created between the film part **340** and the bending part **130b**, and the insulation distance and creepage distance between the bottom part **300a** of the frame member and the end of the protrusion part **110p** of the battery cell **110** may be insufficient. Therefore, according to the present embodiment, the insulation performance can be improved.

(43) FIG. 12 is a modified embodiment of FIG. 6, which is a cross-sectional view taken along the xz plane which is the longitudinal direction of the battery cell stack in FIG. 3.

(44) Referring to FIG. 12, the battery module according to the present embodiment includes a film part **340** that is connected to the pad part **320** and protrudes toward the bus bar frame **130**. The pad part **320** may be located on the second portion **300a2**, and the film part **340** may be located on the

first portion **300a1**. At this time, as shown in FIG. **12**, the film part **340** may be formed so as to be in close contact with a step connecting the first portion **300a1** and the second portion **300a2** of the bottom part **300a** of the module frame. The film unit **340** according to the present embodiment may be integrally formed with the pad part **320** to constitute the insulation reinforcing member **345**, or may be attached to the lower end of the pad part **320**. Since the film part **340** is formed so as to be in close contact with the step, insulation between the protrusion part **110p** of the battery cell **110** and the module frame **300** can be further strengthened.

(45) In addition to the differences described above, all the contents described with reference to FIG. **6** can be applied to the present embodiment.

(46) Meanwhile, one or more battery modules according to an exemplary embodiment of the present disclosure may be packaged in a pack case to form a battery pack.

(47) The above-mentioned battery module and a battery pack including the same may be applied to various devices. These devices may be applied to vehicles such as an electric bicycle, an electric vehicle, a hybrid vehicle, but the present disclosure is not limited thereto and can be applied to various devices that can use the battery module and the battery pack including the same, which also belongs to the scope of the present disclosure.

(48) Although the invention has been shown and described above with reference to the preferred embodiments, the scope of the present disclosure is not limited thereto, and numerous other modifications and improvements made by those skilled in the art using the basic principles of the invention described in the appended claims will fall within the spirit and scope of the present disclosure.

DESCRIPTION OF REFERENCE NUMERALS

(49) **110p**: protrusion part **140**: cell block **320**: pad part **340**: film part **340p**: protruding film part **345**: insulation reinforcing member

Claims

1. A battery module comprising: a battery cell stack in which a plurality of battery cells are stacked in a first direction, a bus bar frame coupled to at least one of a front end and a rear end of the battery cell stack, the front end and the rear end of the battery cell stack spaced apart in a second direction, a frame member that houses a cell block, the cell block including the battery cell stack and the bus bar frame, a pad part located at one end of a bottom part of the frame member, and a film part located between the pad part and the bus bar frame in the second direction.
2. The battery module of claim 1, wherein the bottom part of the frame member comprises a first portion and a second portion, the first portion located at an edge with respect to a longitudinal direction of a battery cell of the plurality of battery cells, the second portion located inside the first portion, and wherein a thickness of the first portion is less than a thickness of the second portion.
3. The battery module of claim 2, further comprising a thermal conductive resin layer located between the second portion and the battery cell stack, wherein the pad part is located between the thermal conductive resin layer and the first portion.
4. The battery module of claim 3, wherein the pad part is located on the second portion, and the film part is located on the first portion.
5. The battery module of claim 2, wherein the film part is formed so as to be in contact with a step portion connecting the first portion and the second portion of the bottom part of the frame member.
6. The battery module of claim 2, wherein the frame member is opened on both sides facing each other with respect to the longitudinal direction of the battery cell, the bus bar frame is connected to the battery cell stack on the opened sides of the frame member, wherein the bus bar frame comprises a main frame arranged perpendicular to the longitudinal direction of the battery cell and a bending part extending from a lower part of the main frame, and wherein the bending part is located on the first portion of the bottom part of the frame member.

7. The battery module of claim 6, wherein the film part extends between the bending part of the bus bar frame and the second portion of the bottom part of the frame member in the second direction.
8. The battery module of claim 7, wherein the film part further comprises a protrusion film part located between the bending part of the bus bar frame and the first portion of the bottom part of the frame member in the second direction.
9. The battery module of claim 6, wherein a sum of a thickness of the bending part and a thickness of the first portion is less than a thickness of the second portion.
10. The battery module of claim 6, each battery cell of the plurality of battery cells comprises a protrusion part formed in a width direction, and the protrusion part is located on the bending part.
11. The battery module of claim 1, wherein the pad part and the film part are formed.
12. The battery module of claim 1, wherein the film part is attached to a lower end of the pad part.
13. The battery module of claim 1, further comprising end plates each respectively coupled to opened sides of the frame member, and wherein the opened sides of the frame member face each other with respect to a longitudinal direction of a battery cell of the plurality of battery cells.
14. The battery module of claim 1, wherein the frame member comprises a module frame that houses the cell block and has an opened upper part, and an upper plate that covers the cell block on the module frame, and wherein the module frame comprises a bottom part of the frame member and two side surface parts facing each other.
15. A battery pack comprising the battery module of claim 1.
16. The battery module of claim 1, wherein a thickness of the pad part is greater than a thickness of the film part.
17. The battery module of claim 1, wherein the bus bar frame comprises a bus bar, and wherein electrode leads extend through slits in the bus bar.
18. The battery module of claim 1, wherein the film part is connected to the pad part.
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